Let $A = (a, a+\delta) \cup \left(a+\frac{1}{k}, a+\frac{1}{k}+\delta\right) \cdots \cup$

$\cdots \cup \left(a+\frac{m-1}{k}, a+\frac{m-1}{k}+\delta\right),$

that is, $A = \bigcup_{r=0}^{m-1} \underbrace{\left(a+\frac{r}{k}, a+\frac{r}{k}+\delta\right)}_{I_r}$

where, for the I_r's to be disjoint,

$(*)\ \frac{m-1}{k}+\delta \leq 1 \quad (**)\ \delta \leq \frac{1}{k}$

Note that

$L(I_r + I_s - k I_t) =$

$(k-2)a + \frac{1}{k}(r+s-kt) - k\delta$

$R(I_r + I_s - k I_t) =$

$= (k-2)a + \frac{1}{k}(r+s-kt) + 2\delta$

Since $\frac{1}{K} \cdot K \cdot t$ is an integer, this sits in between 2 integers

$$\Leftrightarrow \Big(\underbrace{(K-2)a + \frac{r+s}{K} - K\delta}_{L'_{rs}}, \underbrace{(K-2)a + \frac{r+s}{K} + 2\delta}_{R'_{rs}} \Big)$$

does. In order to maximize m, we set $(K-2)a - K\delta \in \mathbb{Z}$, $\left(a = \frac{K\delta}{K-2} \bmod 1\right)$

Then, for example, if

$$\frac{2(m-1)}{K} + (K+2)\delta = R'_{(m-1,m-1)} - L'_{0,0} \le 1,$$

then all (L_{rst}, R_{rst}) sit between 2 integers $\Rightarrow$ We may set

$$\delta := \left(1 - \frac{2(m-1)}{K}\right) \cdot \frac{1}{K+2} =$$

$$= \frac{k-2m+2}{k(k+2)} < \frac{k+2}{k(k+2)} < \frac{1}{k}$$ (**) ✓

Now we maximize over m $\varphi(m)$

the value $m \cdot \delta = \frac{1}{k(k+2)} \overbrace{(k-2m+2)\, m}$.

We will check later condition (*) so that all I_n's are disjoint so this is the actual value of $\mu(A)$.

$\varphi(m)$ is an upside down parabola with roots at 0 and $\frac{k+2}{2}$ $\Rightarrow$ the max is at $m = \left\lceil \frac{k+2}{4} \right\rceil$ or $\left\lfloor \frac{k+2}{4} \right\rfloor$, where: $\left| m - \frac{k+2}{4} \right| \leq \frac{1}{2}$. Since

$\varphi\left(\frac{k+2}{4} + x\right) = \varphi\left(\frac{k+2}{4} - x\right)$ and

φ is decreasing in $\left[\frac{k+2}{4}, +\infty\right)$,

we may assume $m_0 = \frac{k+2}{4} + \varepsilon$,

$\varepsilon \in [0, 1/2]$. and

$$\varphi\left(\frac{k+2}{4}\right) - \varphi(m) \leq \varphi\left(\frac{k+2}{4}\right) - \varphi\left(\frac{k+2}{4} + \frac{1}{2}\right) =$$

$$= \varphi\left(\frac{k+2}{4}\right) - \varphi\left(\frac{k+4}{4}\right) =$$

$$= \frac{1}{8}\Big((2k+4-k-2)(k+2)$$

$$+ (2k+4-k-4)(k+4) \Big) =$$

$$= \frac{(k+2)^2 - k^2 - 4k}{8} = \frac{4-2k}{8} < 1$$

therefore, since $\varphi(m_0)$ is an integer

and $\varphi(m) \le \varphi\left(\frac{k+2}{4}\right) \le \varphi(m) + 1$,

$$\varphi(m) = \left\lfloor \varphi\left(\frac{k+2}{4}\right) \right\rfloor, \text{ and}$$

$$m_0 \cdot \delta = \frac{\left\lfloor \frac{k+2}{4}\left(k+2-\frac{k+2}{2}\right) \right\rfloor}{k(k+2)} =$$

$$= \frac{\left\lfloor \frac{1}{8}(k+2)^2 \right\rfloor}{k(k+2)} \overset{(?)}{>} \frac{1}{8}.$$

It only remains to prove that (*) $\frac{m-1}{k}+\delta\le 1.$

(**) $< \frac{m-1}{k}+\frac{1}{k}=\frac{m}{k}\le\frac{\frac{k+2}{4}+1}{k}=$

$=\frac{k+6}{4k}$. If >1, $k+6>4k\Rightarrow$

$\Rightarrow k<3$, against our assumption.